

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
KHWOPA COLLEGE OF ENGINEERING
Libali, Bhaktapur

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Department of Computer and Electronics Engineering

A REPORT ON

Project Title Goes Here

Submitted in partial fulfillment of the requirements for the degree of

BACHELOR OF COMPUTER ENGINEERING

Submitted by:

Member One Name	KCE0XXBCT0XX
Member Two Name	KCE0XXBCT0XX
Member Three Name	KCE0XXBCT0XX
Member Four Name	KCE0XXBCT0XX

Under the Supervision of

Er. Supervisor Name
Lecturer
Department of Computer and Electronics Engineering
KHWOPA COLLEGE OF ENGINEERING

Libali, Bhaktapur, Nepal

Month, Year

CERTIFICATE OF APPROVAL

This is to certify that this minor project work entitled “Project Title Goes Here” submitted by Member One Name (KCE0XXBCT0XX), Member Two Name (KCE0XXBCT0XX), Member Three Name (KCE0XXBCT0XX) and Member Four Name (KCE0XXBCT0XX) has been examined and accepted as the partial fulfillment of the requirements for the degree of BACHELOR OF COMPUTER ENGINEERING. The project was carried out under special supervision and within the time frame prescribed by the syllabus.

Er. External Examiner Name
External Examiner
Senior Lecturer
Department of Computer and
Electronics Engineering
External College Name

Er. Supervisor Name
Project Supervisor
Lecturer
Department of Computer and
Electronics Engineering
KHWOPA COLLEGE OF
ENGINEERING

Er. HOD Name
Head of Department
Department of Computer and Electronics
Engineering
KHWOPA COLLEGE OF ENGINEERING

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Head of Department

Department of Computer and Electronics Engineering

KHWOPA COLLEGE OF ENGINEERING

Libali, Bhaktapur, Nepal.

ACKNOWLEDGEMENT

Member One Name KCE0XXBCT0XX
Member Two Name KCE0XXBCT0XX
Member Three Name KCE0XXBCT0XX
Member Four Name KCE0XXBCT0XX

ABSTRACT

Keywords:

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LIST OF SYMBOLS AND ABBREVIATION

API	Application Programming Interface
CNN	Convolutional Neural Network
DFD	Data Flow Diagram
GPU	Graphics Processing Unit
RAM	Random Access Memory
UI	User Interface

CHAPTER 1

INTRODUCTION

1.1 Background

1.2 Motivation

1.3 Problem Statement

-
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-

1.4 Objective

-

1.5 Scope and Applications

1.5.1 Scope

1.5.2 Applications

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CHAPTER 3

METHODOLOGY

3.1 System Block Diagram

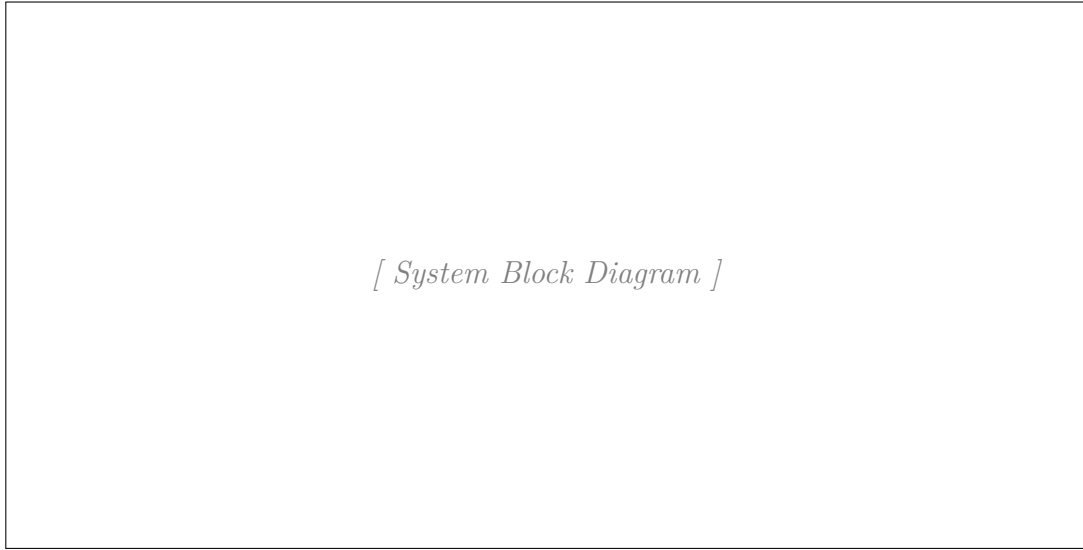


Figure 3.1: System Block Diagram

3.2 Description of Working Flow of System

3.2.1 Dataset Collection

3.2.2 Data Organization

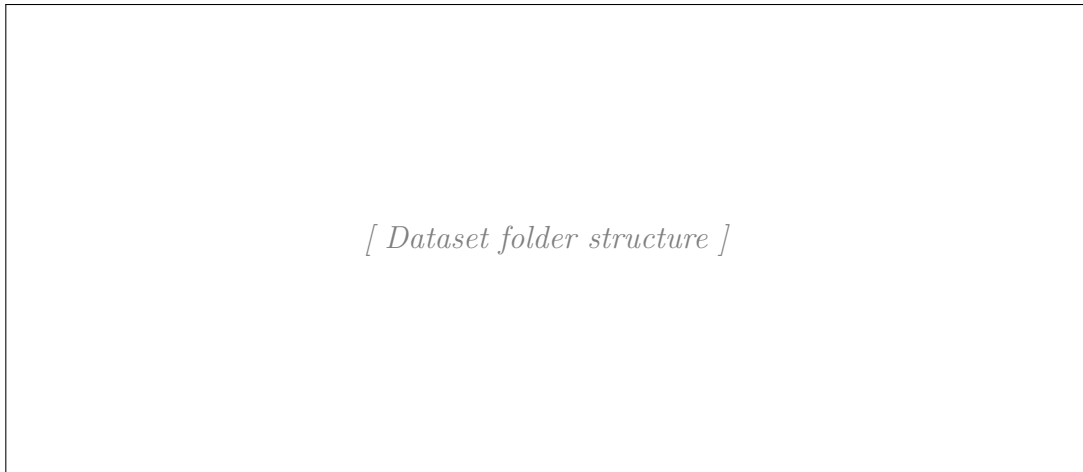


Figure 3.2: Dataset folder structure.

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3.2.6.2 System Integration

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3.2.7.2

3.2.7.3

3.2.8 Final Output Generation

3.3 Model Architecture

3.3.1 Architecture

3.3.1.1

[Model 1 Architecture]

3.3.1.2

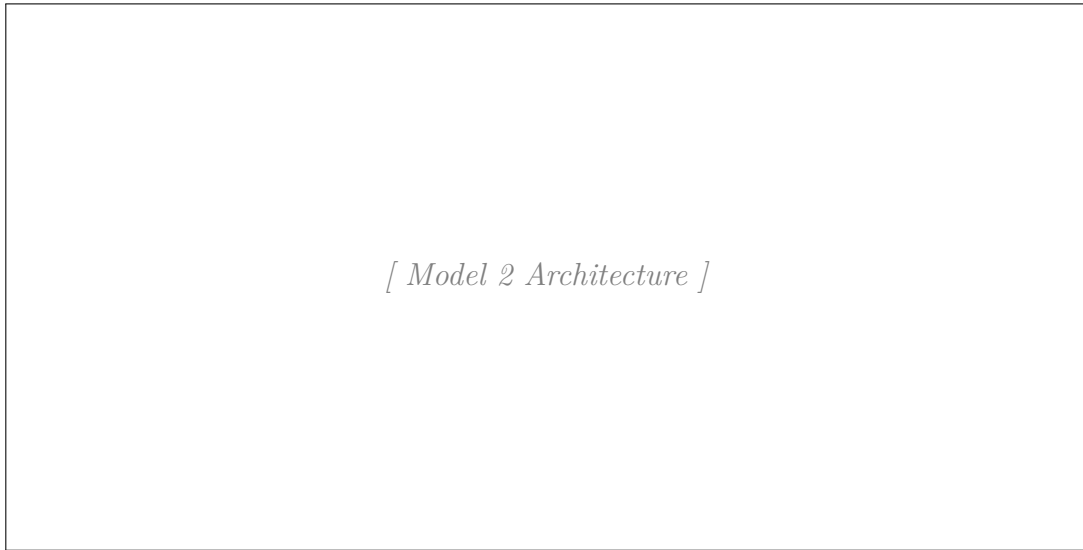


Figure 3.4: Model 2 Architecture

3.3.1.3

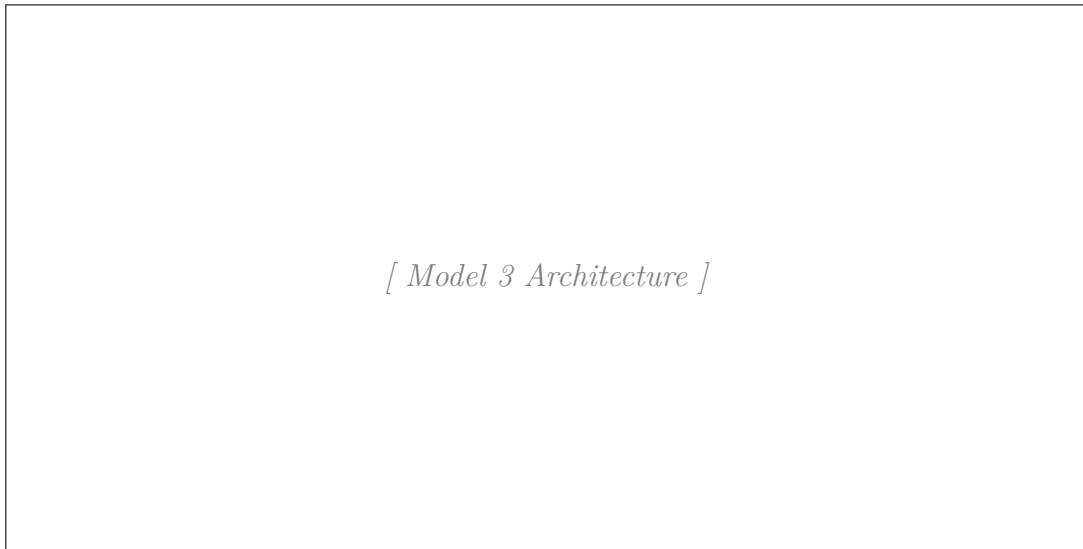


Figure 3.5: Model 3 Architecture

3.3.2 Hyperparameters

3.3.2.1

Table 3.1: Model 1 Hyperparameters

Hyperparameter	Value

3.3.2.2

Table 3.2: Model 2 Hyperparameters

Hyperparameter	Value

3.3.2.3

Table 3.3: Model 3 Hyperparameters

Hyperparameter	Value

3.3.3 Hyperparameter Tuning Process

3.3.3.1

3.3.3.2

3.4 Performance Evaluation Metrics

3.4.1



Figure 3.6: Confusion Matrix

3.4.2

CHAPTER 4

SYSTEM DESIGN

4.1 Requirement Specification

4.1.1 Functional Requirements

-
-
-

4.1.2 Non-Functional Requirements

-
-
-

4.2 Context Diagram

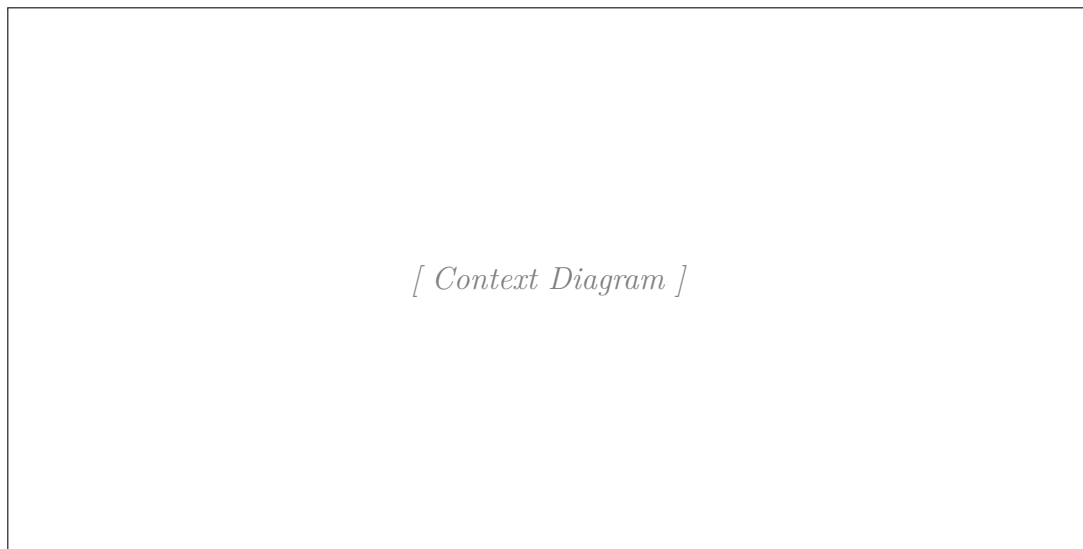


Figure 4.1: Context Diagram

4.3 Data Flow Diagram

4.3.1 DFD Level 0

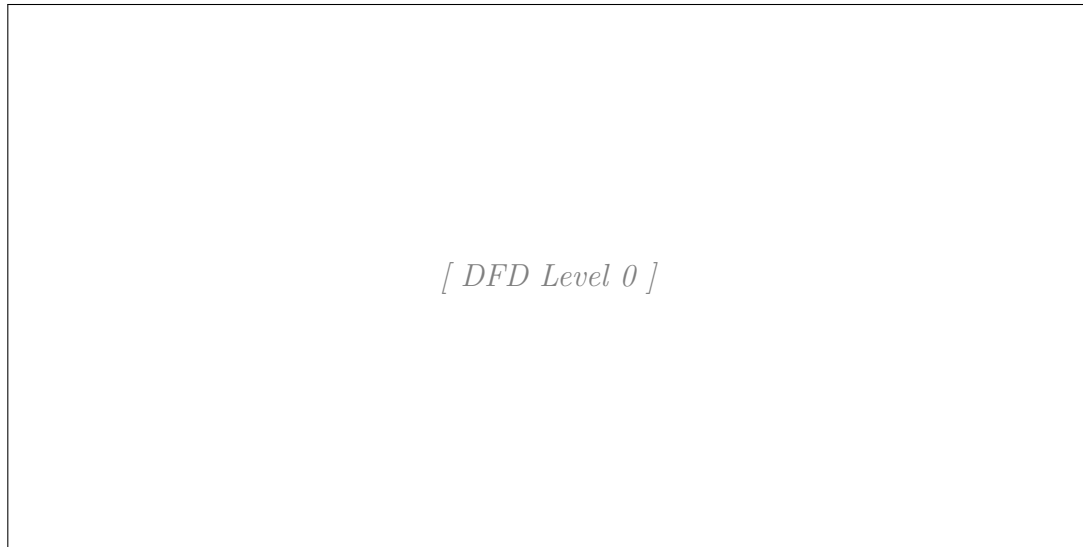


Figure 4.2: DFD Level 0

4.3.2 DFD Level 1



Figure 4.3: DFD Level 1

4.4 Use Case Diagram

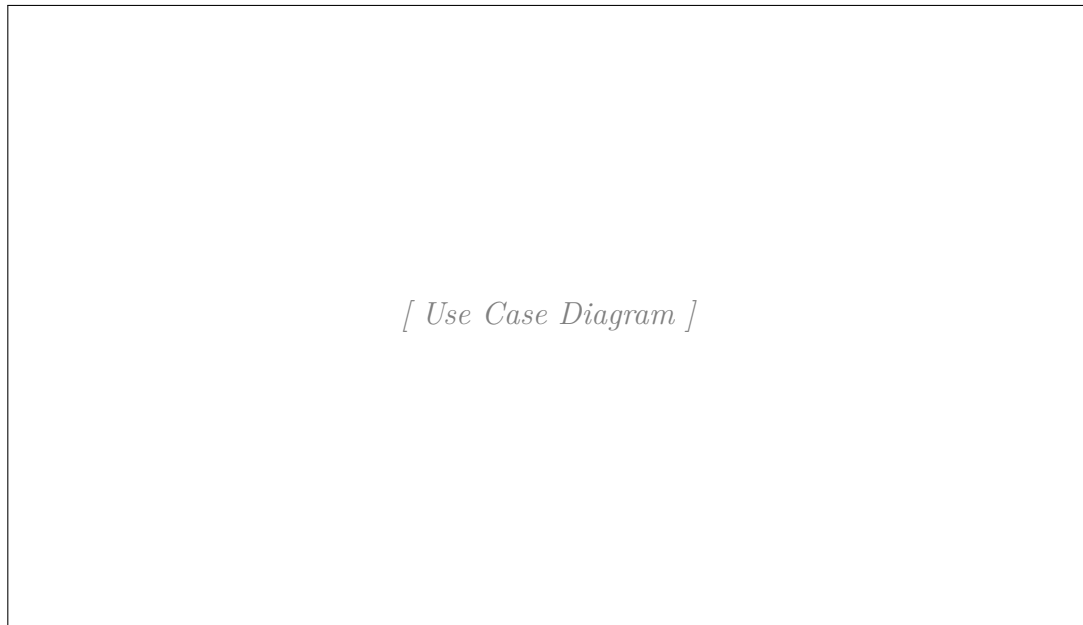


Figure 4.4: Use Case Diagram

4.5 Class Diagram

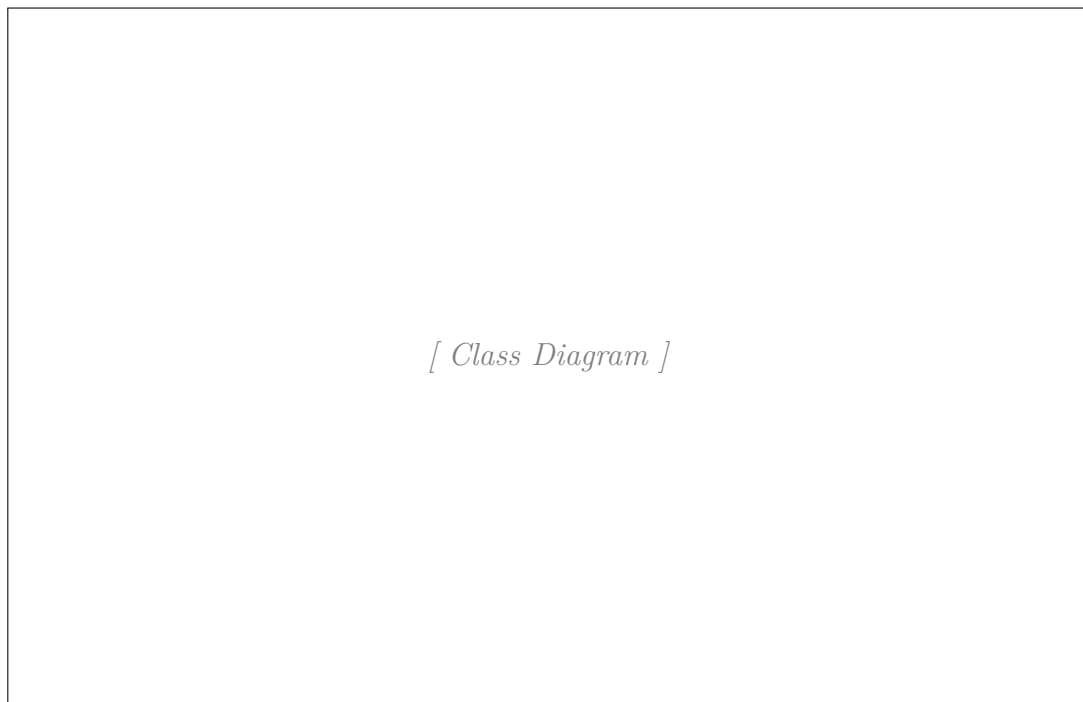


Figure 4.5: Class Diagram

4.6 Sequence Diagram

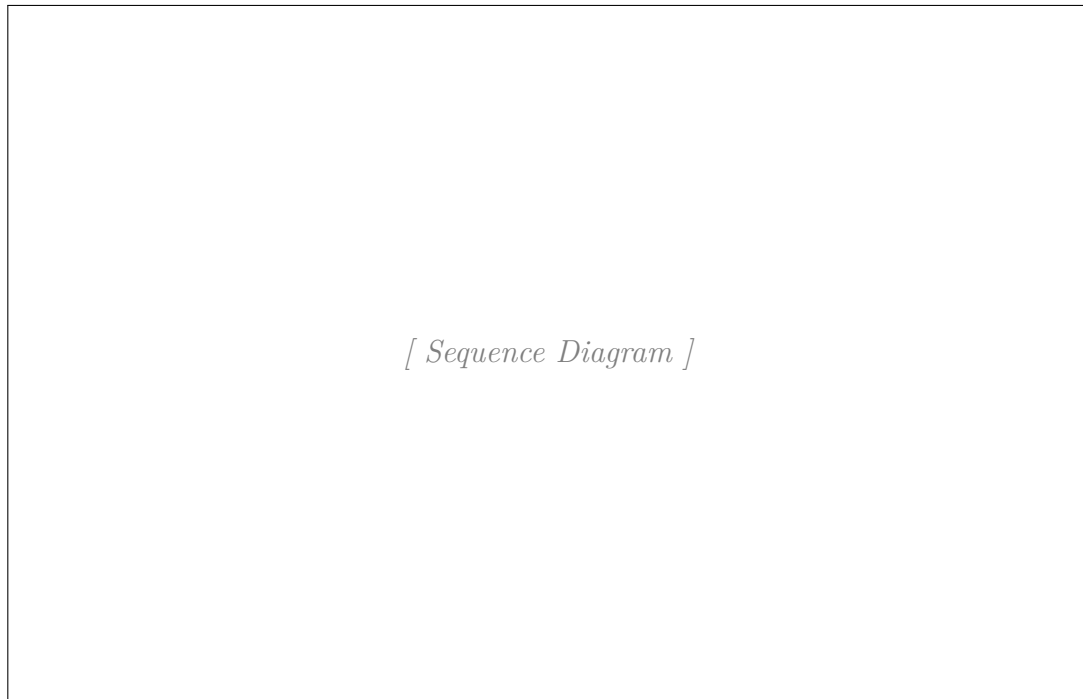


Figure 4.6: Sequence Diagram

4.7 Component Diagram

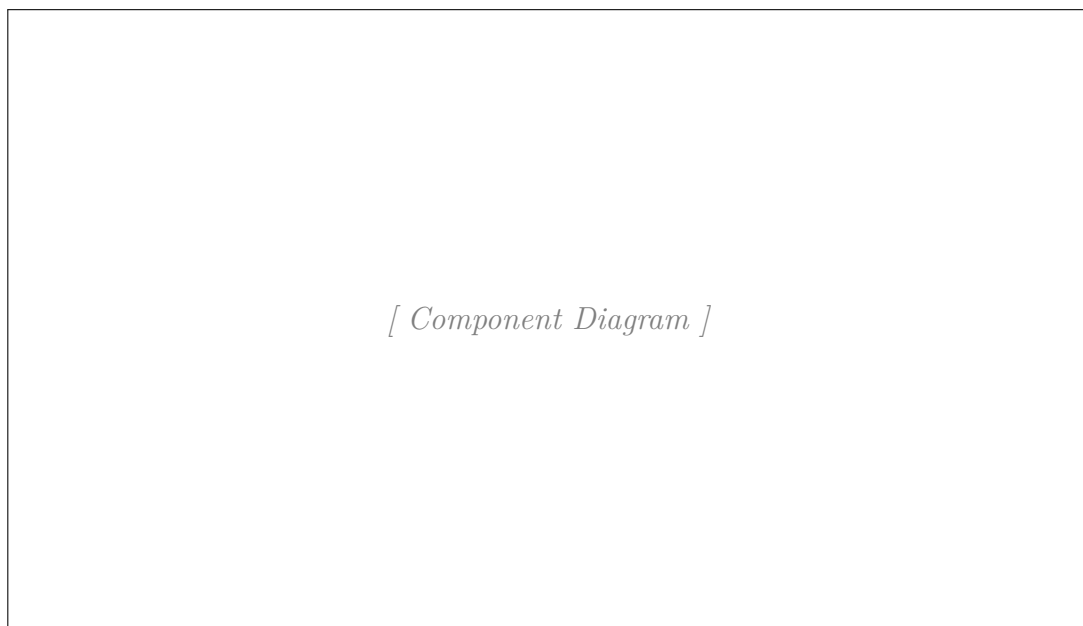


Figure 4.7: Component Diagram

4.8 Deployment Diagram

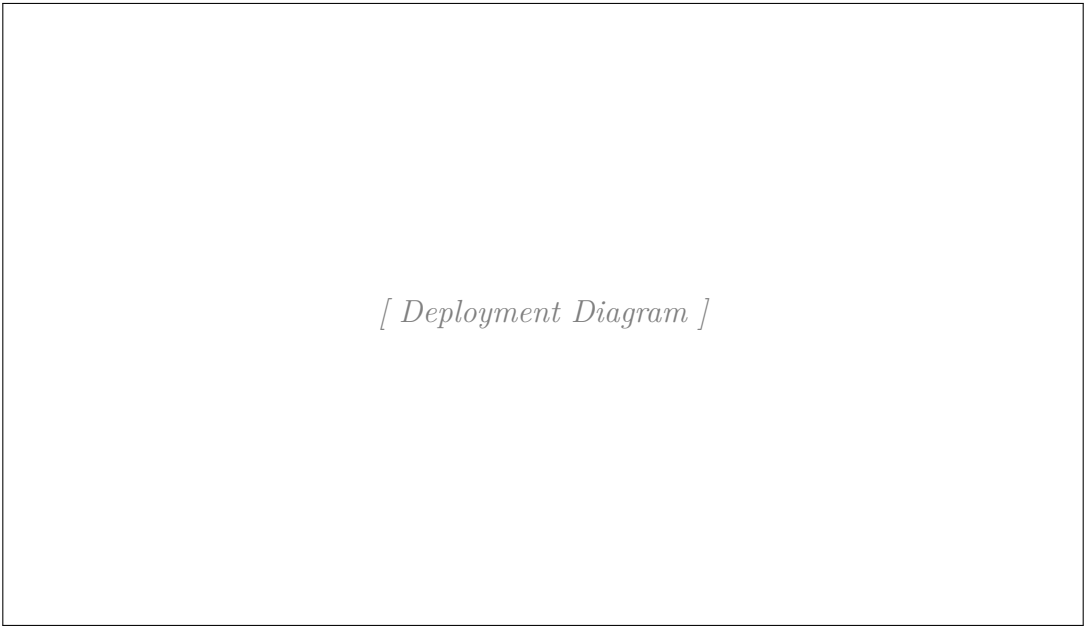


Figure 4.8: Deployment Diagram

4.9 Gantt Chart

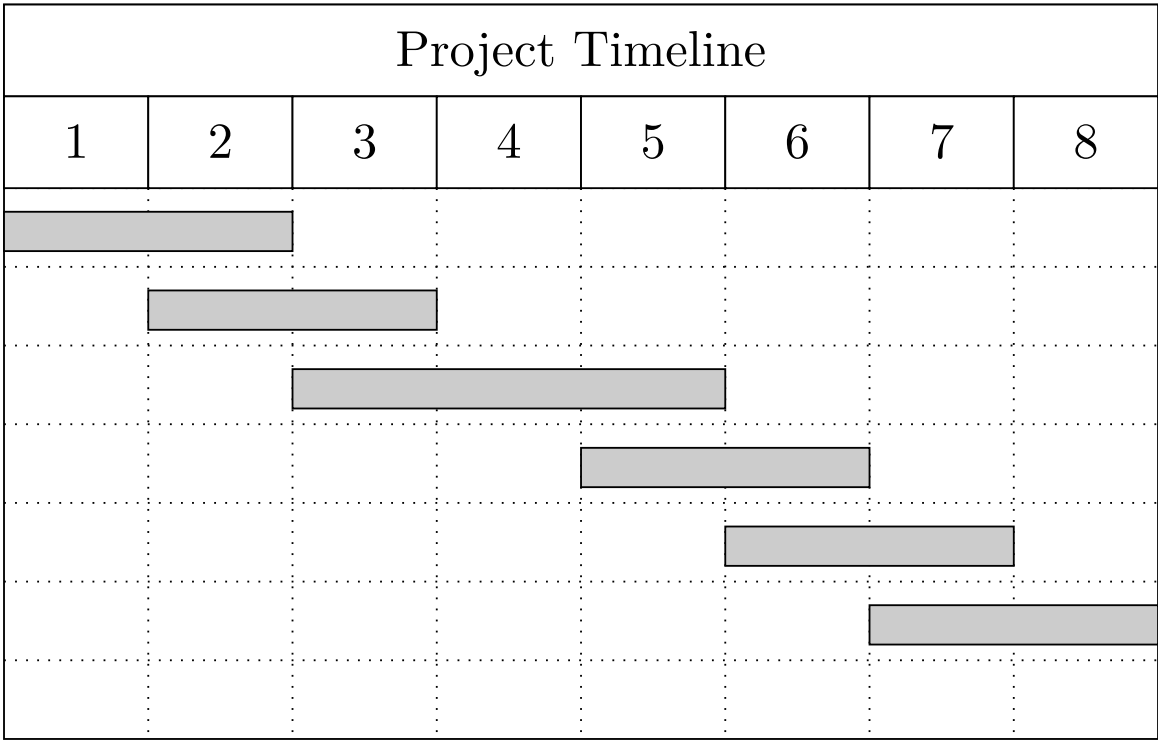


Figure 4.9: Project Gantt Chart

CHAPTER 5

RESULT AND DISCUSSION

5.1 Experimental Setup

5.1.1

5.1.1.1

5.1.1.1.1 Model Training Phases

5.1.1.1.2



Figure 5.1: Model 1 Training and Validation Loss and Accuracy Curves

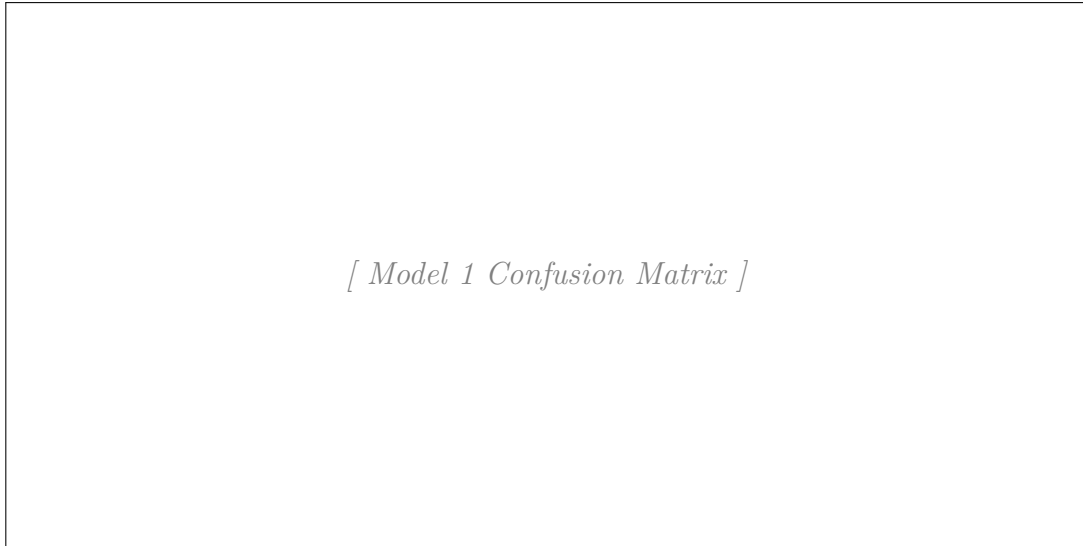


Figure 5.2: Model 1 Confusion Matrix

5.1.1.2

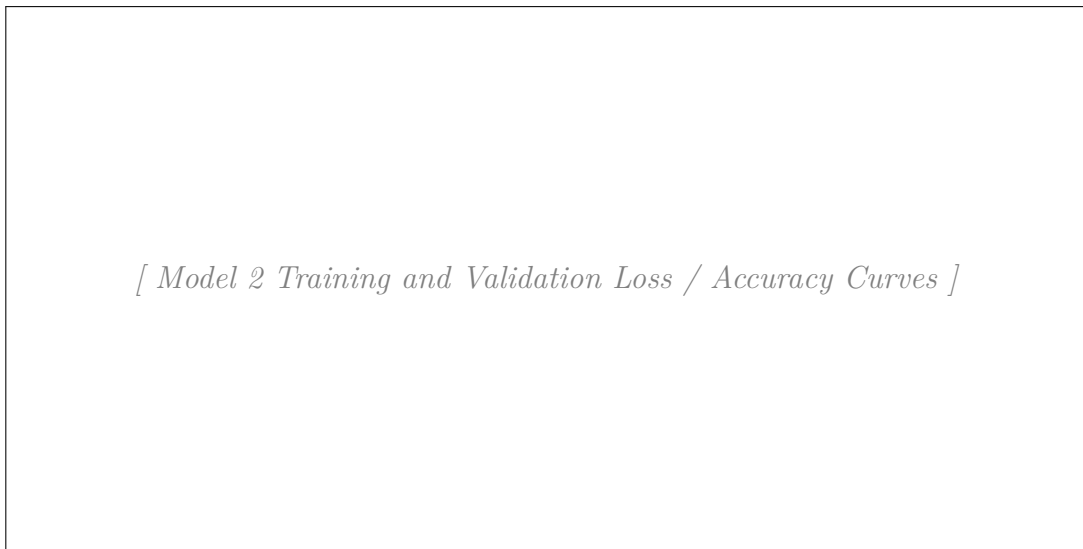


Figure 5.3: Model 2 Training and Validation Loss and Accuracy Curves

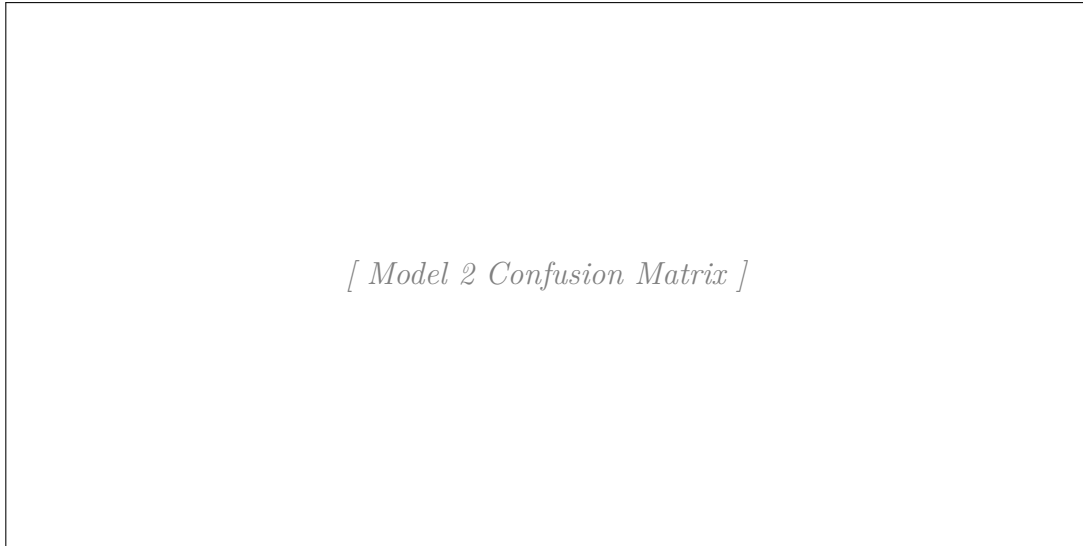


Figure 5.4: Model 2 Confusion Matrix

5.1.1.3 ROC Curve Analysis

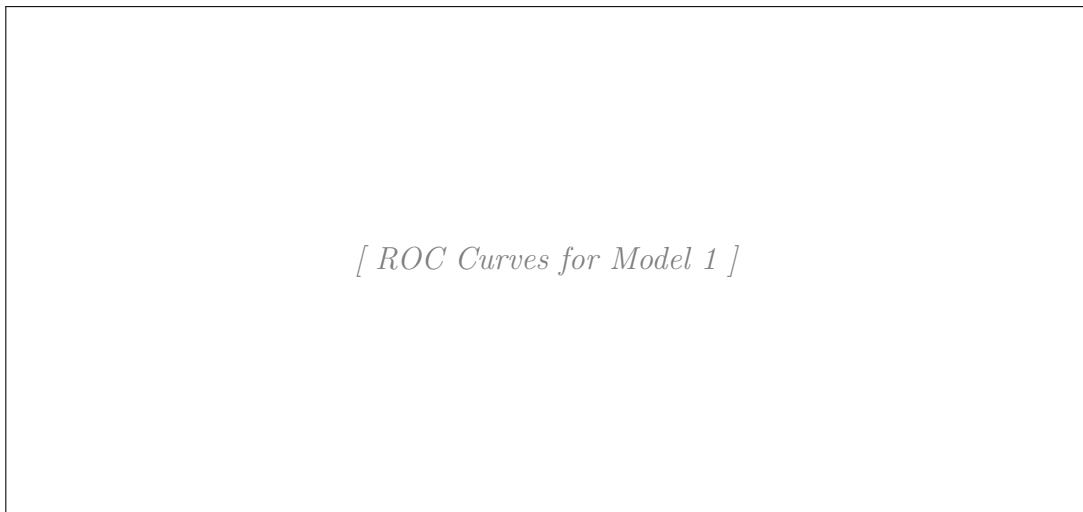


Figure 5.5: ROC Curves for Model 1

5.1.1.5 Training Behavior Analysis

5.1.1.5.1 Convergence Speed

5.1.1.5.2 Learning Rate Sensitivity

5.1.1.5.3 Generalization Gap

5.1.1.5.4 Computational Efficiency

5.1.2

5.1.2.1 Model Training Phases

5.1.2.1.1 Phase I

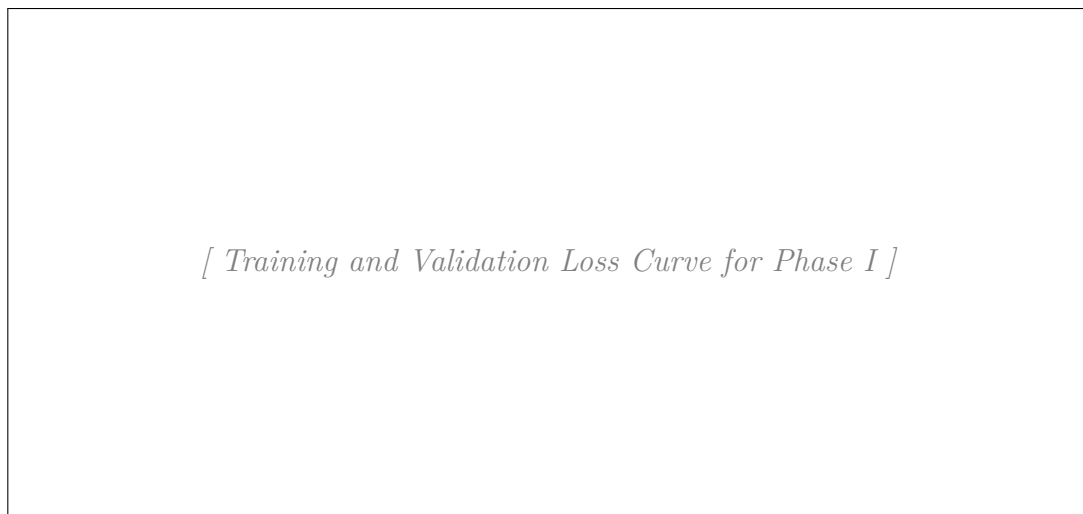


Figure 5.7: Training and Validation Loss Curve for Phase I

5.1.2.1.2 Phase II

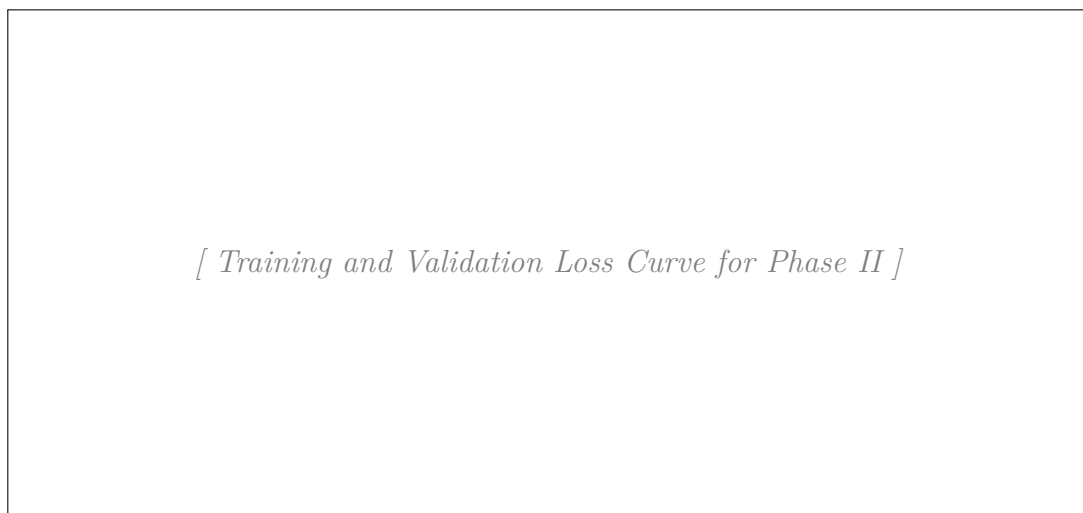


Figure 5.8: Training and Validation Loss Curve for Phase II

5.1.2.1.3 Phase III

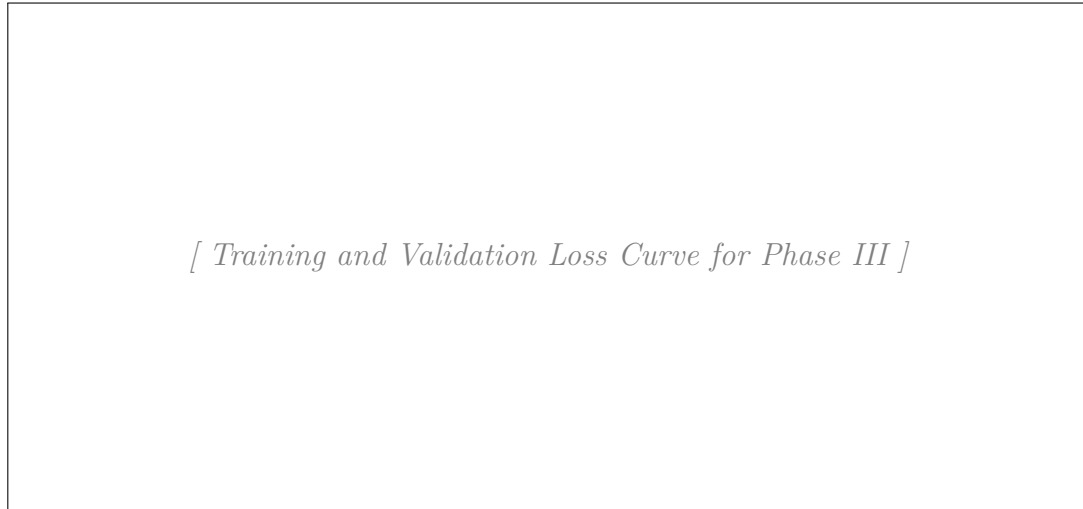


Figure 5.9: Training and Validation Loss Curve for Phase III

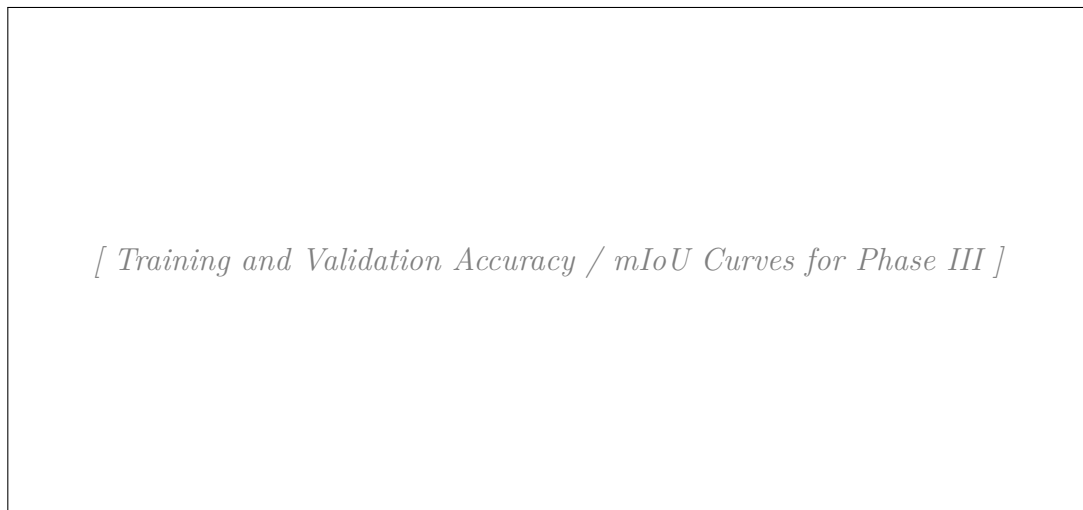


Figure 5.10: Training and Validation Accuracy and mIoU Curves for Phase III

5.1.2.2 Model Behaviour Across Dataset Splits

5.1.2.3 Final Model Configuration

5.2 Software Development Approach

5.2.1 Agile Development Methodology



Figure 5.11: Agile Software Development Cycle

5.2.2 Agile Sprints

5.2.2.1 Sprint 1: Research and Planning

5.2.2.2 Sprint 2: Data Collection and Preprocessing

5.2.2.3 Sprint 3:

5.2.2.4 Sprint 4:

5.2.2.5 Sprint 5:

5.2.3 Sprint Summary

Table 5.2: Sprint Summary

Sprint	Focus	Activities	Deliverable
1			
2			
3			
4			
5			

5.3 Model Deployment

5.3.1 Frontend Development

5.3.2 Backend Development

5.3.3 Model Integration

5.3.3.1 API Integration

- Request:
- Response:

CHAPTER 6

CONCLUSION AND RECOMMENDATION

6.1 Conclusion

6.2 Recommendations

-
-
-

6.3 Limitations and Future Work

-
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BIBLIOGRAPHY

[1]

[2]

[3]

Appendix

A. User Interface Screenshots

A1.

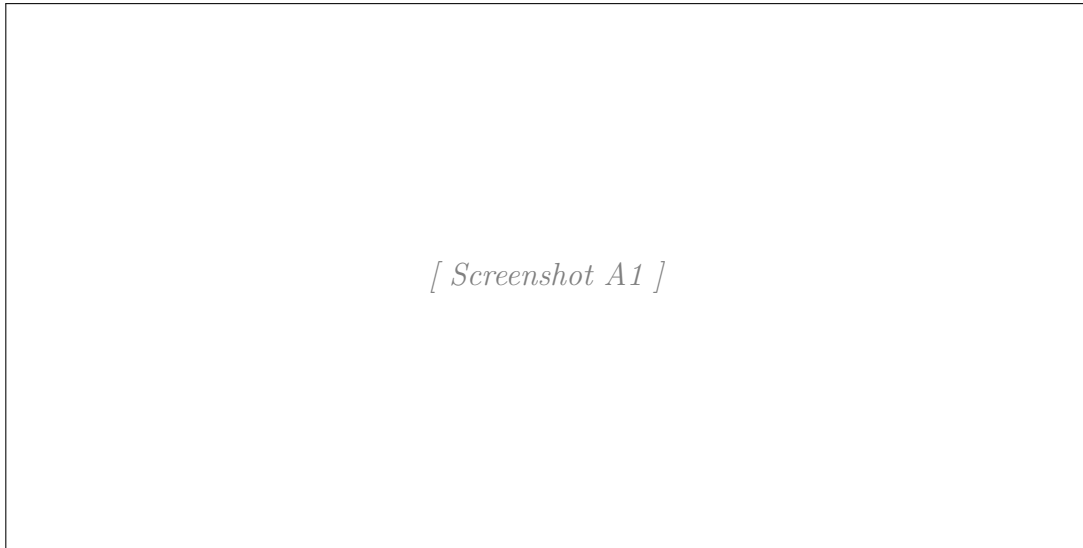


Figure 6.1

A2.

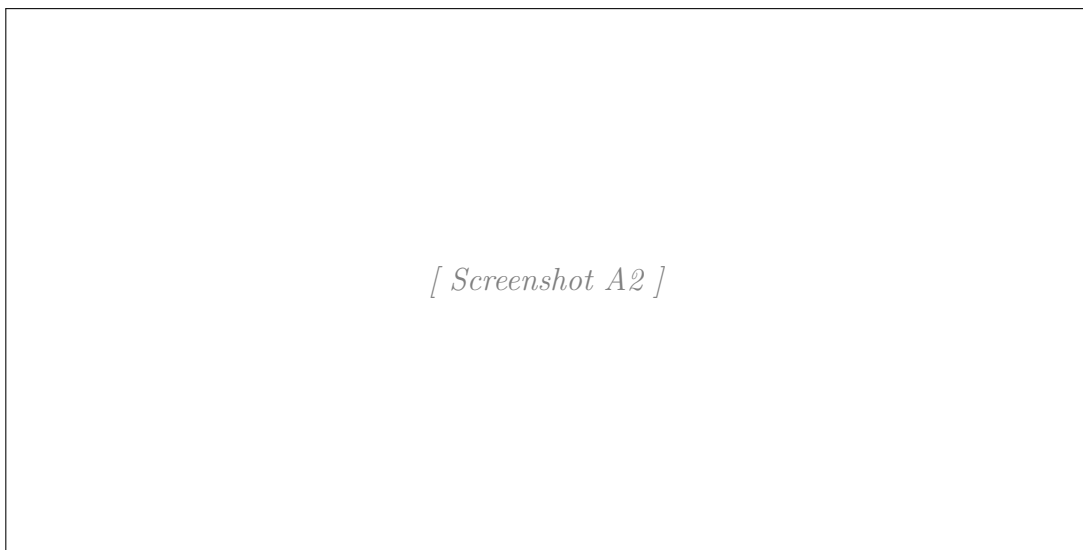


Figure 6.2

A3.

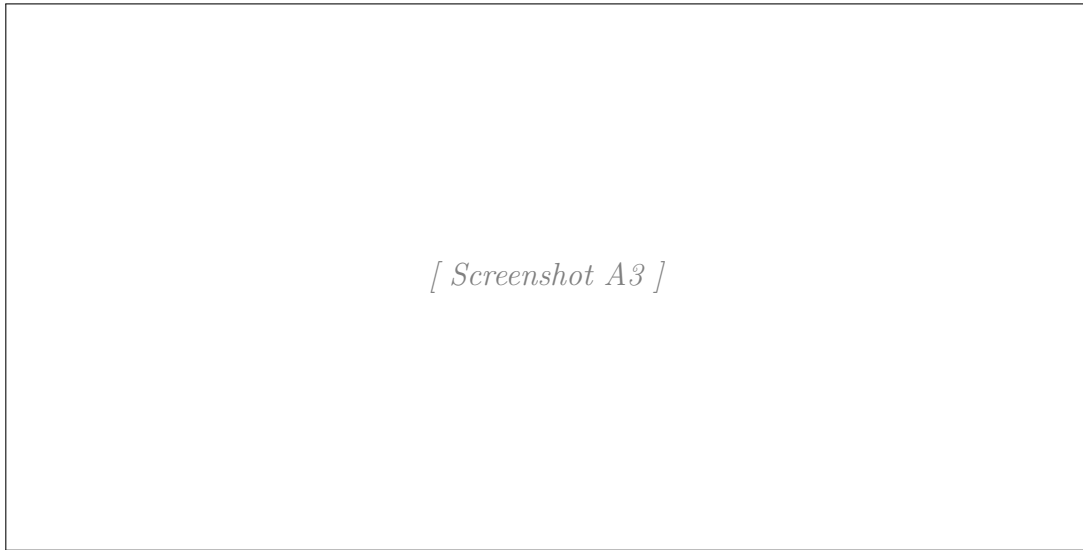


Figure 6.3

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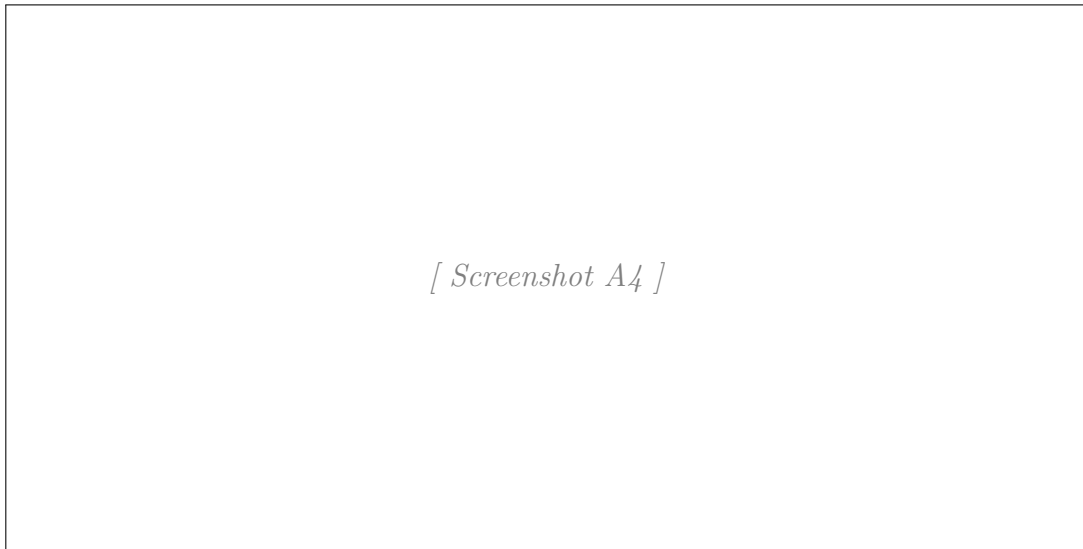
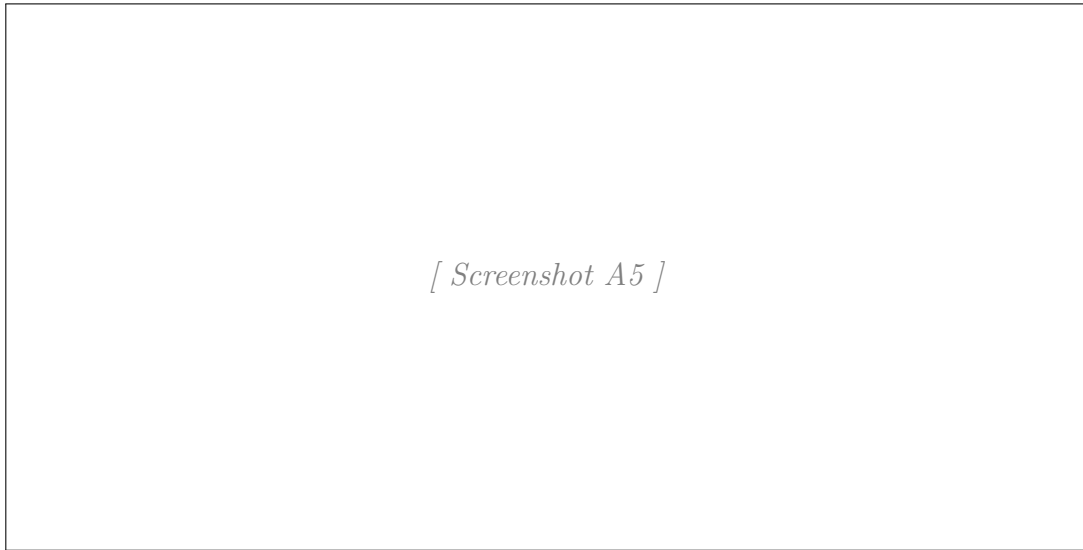


Figure 6.4

A5.



[Screenshot A5]

Figure 6.5